White Dog Cafe

"Business, I learned, is about relationships. Money is simply a tool. What matters most are the relationships with everyone we buy from, sell to, and work with."

Judy Wicks, Founder
White Dog Café
"Good Morning, Beautiful Business"
Chelsea Green Publishing, 2013

Penn Engineering

Leaders & Managers

Creators & Equilibrators

Creative Destruction

"The entrepreneurial/leadership function is creative – not 'equilibrating.' Entrepreneurship is a process of creative destruction; one that incessantly revolutionizes the economic system from within, incessantly destroying the old one, incessantly creating a new one. This process of Creative Destruction is the essential fact about capitalism."

- Joseph Schumpeter, 1934

Development of a Manager
(a.k.a. Equilibrator, Steward)

- **Peaceful life**
  - Easy adjustments along the way
  - Expectations = realizations
  - Rely on order and stability
  - Uncomfortable with disorder
- **Develop through socialization**

… prepares them to guide institutions and maintain balance.

Development of a Leader
(a.k.a. Creator)

- **Expectations of achievement**
  - Sense of being special, self-awareness
  - Self-reliant
  - Desire for autonomy
- **Develop through personal mastery**

… driven to do great works and create change.

Sense of Self

- **Leaders**
  - Do not depend on memberships
  - Do not “belong” to organizations
  - Desire to upset the applecart
- **Managers**
  - Seek positive attachments
  - Identify with the existing order
  - Duty and responsibility

Relationships

- **Leaders**
  - Comfortable alone
  - Empathetic
  - Intensive attachments
  - Subordinates:
    - Have strong feelings
    - May be turbulent
- **Managers**
  - Seek collaborators
  - Linked to organization
  - Moderate attachments
  - Subordinates:
    - Detached feelings
    - May feel manipulated

Leaders have more in common with artists, scientists, engineers and other creative thinkers than they do with managers.

- Abraham Zaleznik, 1992

High-Tech Ventures Need Both

- **Leaders**
  - Visionaries
  - Active: create customer "want"
  - Establish the culture
  - Open issues and new options
  - Driven by pursuit of new opportunity
  - Ignore "legitimate" paths
  - Inspire the team to create
    - Intrinsic motivator
- **Managers**
  - Execute the vision
  - Reactive: satisfy customer "need"
  - Perpetuate the culture
  - Narrow choices and limit options
  - Driven by need to maintain order
  - Success with organization
  - Inspire the team to succeed
    - Extrinsic & intrinsic motivator

"The entrepreneurial function may be and often is filled cooperatively – in many cases, therefore, it is difficult or even impossible to name an individual that acts as "the entrepreneur."

- Joseph Schumpeter, 1949

Palm Computing, Inc.

- **A leader/manager partnership**
  - **Jeff Hawkins**
    - BSEE, Cornell University
    - Insatiable curiosity about pattern recognition
  - **Donna Dubinsky**
    - MBA, Harvard Business School
    - Brought order to Apple's inventory systems

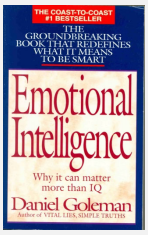
Bridging the Gap

Leaders (Creators) & Managers (Stewards)

- **Keep creators around…**
  - Intrinsically motivate your creators
  - Grow up but stay young!
- **Balance influence**
  - Results-oriented managers vs technical visionaries
  - Encourage bridging communication
  - Encourage debate
- **Structure the innovation process, but…**
  - Avoid bureaucratic awfulness
  - Keep the spirit and energy alive!

What Makes a Leader?

- **Entry Level Requirements**
  - Intelligence
  - Technical skills
- **Emotional IQ ("EQ")**



… Daniel Goleman, "Emotional Intelligence"
Bantam Books, NY

Emotional IQ

- **Self-Management Skills**
  - Self-Awareness
  - Self-Regulation
  - Motivation
- **Relationship Management Skills**
  - Empathy
  - Social Skills

Self-Management Skill #1:

Self-Awareness

- **Self-Confidence**
  - Recognize core competencies
  - Know when to ask for help
  - Listen
- **Underlying Personal Ethics**
  - Decisions mesh with values
  - Pursue “shoulds” not “wants”
  - Sets the norms. Demonstrates the values.
  - Attracts and keeps the right people

Self-Management Skill #2:

Self-Regulation

- Control inner emotions
- Comfortable with ambiguity
- Flexible
- **Creates an environment of trust**
- Attracts talented people

Self-Management Skill #3:

Motivation

- **Passion for work itself**
  - Achieving for the sake of achievement
- **Restless with the status quo**
  - Looking for a better way
- **Unflagging optimism**
  … but beware of cognizant bias!

Relationship-Management Skill #1:

Empathy

- Understanding and considering others' feelings
- Mentoring relationships
- **Being an intrinsic motivator**

Relationship-Management Skill #2:

Social Skills

- **Relationships beyond job roles**
  - Ability to build rapport with all kinds of people
- **Building networks**
  - No leader is an island

Leadership: it is both a dynamic competence and a relational process, which require specific skills and capabilities for envisioning, for setting aspirations, and for motivating others.

Recap…

17 case studies + 50 articles + 2 videos + 13 essays + 10 quizzes + 1 problem set + 4 guest questions =

"Ridiculous amount of work. Worth it, though."

EAS545 Student, Fall 2013

Louis Agassiz

- **Leaders Look at the Fish!**
  - Never be content with the facts on hand
  - See what others don't see
  - See how little you saw before
  - Don't settle for the easy answers
- **Confident Leaders See and Listen!**

"The doors of wisdom are never shut" ... Benjamin Franklin

Peter Drucker

"… the world's greatest management thinker."

- **4 Requirements of Entrepreneurial Management**
  - Market-driven focus
  - Financial foresight
  - Build a top management team early
  - Objective founder's role

Hewlett Packard

- **Bill Hewlett & David Packard:**
  - First Who… Then What
  - Complementary skills
- **Prof. Fred Terman, mentor**
  - Guidance, support, credibility, contacts
- **Intrinsic motivation**
  - MBWA
  - Trust, respect, open communications
- **Bootstrap finance**
  - Avoided debt: financed growth with earnings
  - Diligent cash management

Vermeer Technologies

- **Charles Ferguson**
  - Inspirational
  - Well connected
  - Knew the market
  - Focused vision
  - Flexible: changed his product

- **Randy Forgaard**
  - Technical competence
  - Pragmatic
  - Empathetic
  - Communicator
  - High status friend

Palm Computing, Inc.

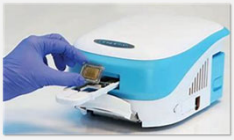
- **Engineers create devices but**
- **Customers buy products**

Dell Computers

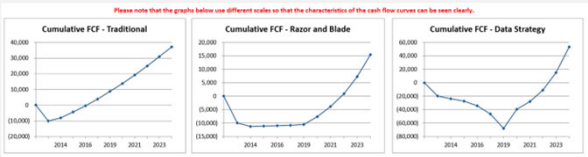
- **Michael Dell**
  - **Vision and focus:**
    - A better way to provide computers
    - Staying close to your customer
    - Importance of cost leadership
  - "You need to get out there and take some risks, or risk the consequences"

GenapSys

DNA Sequencer



- **A disruptive technology**
  - Faster, less costly, smaller footprint
- **Business model decision:**
  - Traditional
  - Razor-and-Blade
  - Data Analytics
- **Time value of money**
  - Upfront capital requirements
  - Timing and amount of net income



NPV:	$37.7M @ 12%	$35.9M @ 15%	$49.8M @ 20%

Sirtris Pharmaceuticals

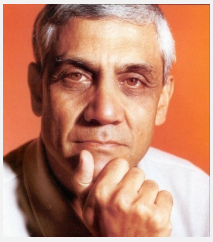
- **Complementary Founders**
  - Dr. David Sinclair, scientist
  - Dr. Christoph Westphal, science & business savvy
    - "A return entrepreneur with a blank check," Rob Stavis, BVP Series A investor
- **World-Class Scientific Advisory Board**
- **Multiple VC rounds at increasing valuations**
  - Value-adding events
- **Operations Strategy**
  - Nutraceuticals?
  - Big-Pharma deal? Now or later?
  - Diversification in-licensing?

MIPS

The Spoiled Startup

- **Lack of leadership**
- **Lack of focused vision**
  - $8MM revenue goal = no vision
- **Robert Miller**
  - Humble, driven, focused
  - Driven by belief in the technology
  - Focused on market-driven products
  - Restored an effective vision & culture

Sun Microsystems



- **Vinod Khosla**
  - A visionary leader with managerial skills
  - A's attract A's
  - Fiscal controls
  - Relentless pursuit of opportunity
    - Crossing the chasm
  - **"It was all based on trust."**

Equity split:

- Contributions?
- Commitment?
- Timing?

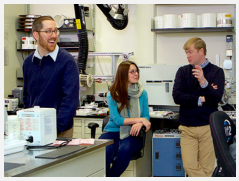
NanoGene Technologies, Inc.

- **What's best for friendships isn't always best for the company:**
  - Differing value to the company.
  - Differing commitment to the company.
  - Differing responsibilities & authority.
- **How many founders are appropriate (& necessary)?**
- **Allocate equity with an eye on the future:**
  - Future dilution by investors.
  - Future dilution by new employees.
  - Future dilution by option pools.
- **Hire ahead of the curve.**

N12 Technologies

Making the Best Better

- **Scaling an informal, loosely run startup**
- **Organizational structure**
  - Horizontal differentiation
  - Vertical differentiation
  - Standardization & formalization
  - Integration



- **Producing, marketing & selling unproven cutting-edge technology**
- **Unattainable 2-year goals**
  - *100X increase in production*
  - *$60M revenues with huge cost reductions*
  - *"Toxic" hired CEO*

Theranos

- **Cognitive biases impair rational decision-making**
- **We all enjoy a good story**
  - Intuitive thinking is effortless and suspends disbelief.
  - Reflective thinking requires an act of will.
  - Intuitive thinking digests a story and suppresses alternatives.
  - Storytelling with emotion effectively conveys visions & ideas, but must have a **rock-solid foundation of honesty and ethics**.
- **Radical transparency improves performance**
  - Accountability is part of a startup's maturation process.
  - Practiced in all communications with all stakeholders.
- **Integrity Is Binary … beware of the slippery slope**

Guest Speakers – *It's about the people*

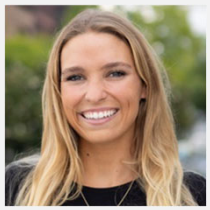
- **Ross Mechanic**
  - Co-Founder & CEO: Maybern
- **Julia O'Mara**
  - Co-Founder & COO: Pickle
- **Melinda Thomas**
  - V.P. of Operations: "NanoGene Technologies, Inc."
  - Co-Founder: Octave Bioscience
- **Tobias Dengel**
  - Co-Founder & CEO: WillowTree Apps
  - President: TELUS Digital Solutions
- **Miranda Wang**
  - Founder & CEO: NovoLoop

Our guests reinforced the principles of EAS5450 - study their classroom slides

Ross Mechanic

- ***Hire slowly. Fire fast.***
  - *Look at the fish when hiring talent.*
  - *Talented people want to work with talented people.*
    - *A's attract A's*
- ***Build bidirectional relationships.***
  - *Investment decisions are 90% emotional*
    - *Leverage relationships.*
    - *Speak with decision makers.*
  - *Build trust rather than going into a meeting to ask for funding*
    - *Don't pitch at the first meeting. Don't give investors a chance to say "no."*
    - *Always fundraising, but never "fundraising."*
    - *Choose investors based on TRUST.*
- ***A lot of the founder's job is story telling***
  - *Learn how to speak, write and to tell a story…*
    - *Storytelling is a massive accelerant.*
    - *Make your story compelling.*
- ***Post-Money SAFEs***
  - *Fast, inexpensive, minimal documentation.*
  - *Until your priced round, nothing dilutes the SAFEs*
    - *Subsequent SAFEs and option pool increases don't dilute SAFEs.*

Julia O'Mara



- ***Developed Pickle before leaving Blackstone.***
  - *Make sure your employer doesn't have IP rights to your idea.*
- ***Embed yourself in customer experience.***
  - *If we build it, they will come: NOT TRUE*
  - *Talked to customers & got feedback.*
  - *Customer retention is really important – focus on customer experience.*
- ***When starting a company, be scrappy and figure out how to do things for free.***
- ***When fund raising, get a warm intro:***
  - *Get a warm intro and meet in person*
  - *Show them your passion*
  - *They're investing in you!*
  - *Once you land your first investor, others will be at your doorstep.*
- ***An in-person culture allows you to learn faster, grow quicker.***

Trust is so important!

Lessons learned: see Julia's slide deck!

Melinda Thomas

a.k.a. Paige Miller

- ***Be careful what you say!***
  - *You never know who's listening.*
- ***Take the time to make the tough decisions ...***
  … they only get tougher later.
  - *Hire them for what they did; compensate them for what they will do.*
  - *Everyone knows you need to move away from "all equal."*
  - *Equal splits won't scale.*
- ***It isn't about the money.***
  - *Be sure compensation isn't the sole or main attraction.*
  - *When good people are available, find a place.*
- ***In year two I began formalizing decision-making processes …***
  - *But did it gradually to avoid stripping their gears.*
- ***Board meetings are not the place to brainstorm ideas.***
  - *Directors want to make decisions tied to the vision, not to debate the vision.*
- ***Don't sit across from each other when negotiating.***

Tobias Dengel

- ***A small team of A+ players can run circles around a giant team of B and C players***
  - *You learn, not so much from what you and the business are working on, but from the people you're working with.*
  - *Know thyself. Be brutally honest with what you're good at, and what you're not good at.*
  - *Surround yourself with people who have capabilities you don't, or don't want to, have.*
- ***Culture is everything***
  - *Why does culture matter? People are primarily driven by normative influence, not objective influence.*
  - *It's how people treat each other.*
- ***MBWA***
  - *What can I do to help you do your job better?*
- ***Business is driven by sales…***
  - *Product/market fit!*
  - *Find good people who can sell.*
  - *Integrating new technology into your life is like building a relationship, it requires **trust** and **value**.*

Lessons learned: see Tobias's slide deck!

Miranda Wang

- ***It's only over when you quit***
- ***You are the biggest investor in your future***
  - *Focus on the solution. The money will follow.*
- ***Be honest with people who take a chance on you***
- ***Build a solution for a worthy problem that won't go away***
  - *Found a company because you're passionate about the work.*
  - *Work with a co-founder who shares your conviction.*
  - *Find where the gaps are, and recruit only the best people.*
- ***It's a real problem when having to pay suppliers faster than revenue income***
- ***Listen to the doubters, and build with your believers***
- ***Be patient and set your life up for long-term success***

Final Thoughts…

Values

Know who you are and, most importantly, behave in a way that is consistent with who you are. At work, don't check who you are at the door.

Egos are necessary for the self-confidence to tackle difficult responsibilities. But, the ability to persist in a leadership role that is fundamentally artificial is short lived.

Great leaders don't draw on their egos, they draw on a sense that comes from within; a moral strength of doing what is "right." This is a leader's self-assurance that drives perseverance and patience.

… John Reed, former CEO Citicorp & NYSE

"If the going is tough and the pressure is on; if reserves of strength have been drained and the summit is still not in sight; then the quality to seek in a person is neither great strength nor quickness of hand, but rather a resolute mind firmly set on its purpose that refuses to let its body slacken or rest."

- Sir Edmund Hillary

EAS5450 Mission:

Providing Creators & Leaders for Technology Ventures

(… but make sure you have a manager)

Savor the Journey…

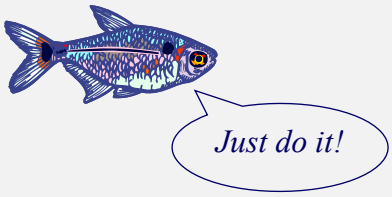
… the Journey is the Reward

Life Lessons: Priorities

- You
- Your family
- Your company
- Your community

…it is the reward.

Engineering Entrepreneurship

Make a difference.

Life begins … where your comfort zone ends!